

Megnath Ramesh

m5ramesh@uwaterloo.ca | +1 (587)-589-5810 | megnath.com | github.com/megalphan

Education

University of Waterloo, Ph.D. in Electrical and Computer Engineering Sept 2020 – August 2025

- **Thesis:** Planning and Replanning Near-optimal Robot Coverage Paths in Partially Unknown Environments
- **Advisors:** Stephen L. Smith and Baris Fidan
- **Key Courses:** Combinatorial Optimization, Computational Geometry, Deep Learning, and Algorithm Design.
- **Awards:** 2x Ontario Graduate Scholarship (\$15k), 3x UW President's Scholarship (\$5k), QEII Graduate Scholarship (\$15k)

University of Alberta, B.Sc. in Electrical Engineering Sept 2013 – May 2018

- **GPA:** 3.8/4.0 - Graduated with Distinction.

Recent Experience

Postdoctoral Fellow (*Vector Faculty Affiliate Researcher*), University of Waterloo September 2025 – Current

- **Advisors:** Stephen L. Smith and Baris Fidan.
- Developing *diffusion models* to plan robot paths for long-horizon coverage of unknown environments.

Ph.D. Researcher, University of Waterloo & Avidbots Corp. Sept 2020 – August 2025

- Developed *coverage planning algorithms* for industrial cleaning robots that achieve near-optimal path lengths and minimize turns.
- Tested algorithms using *real-world robots* to demonstrate applicability while encountering environmental uncertainties.
- *Collaborated with the Avidbots engineering team* towards integrating the algorithms with their autonomy stack.
- *Mentored junior researchers* in devising and implementing solutions to difficult robotics problems.

Graduate Teaching Assistant, University of Waterloo. May 2021 – April 2025

- *Courses:* Algorithm Design and Analysis, Digital Control Applications, and Analog Control Systems.

Application Software Developer (L1 & L2), ScopeAR Jun 2018 – Aug 2020

- Developed features and fixed defects in a cross-platform *augmented reality (AR) video calling* application.
- Worked on a range of features, including *marker-based AR tracking, guest login system, and a pilot cloud-based AR system.*

Junior Robotics Developer, VEERUM Feb 2017 – Apr 2018

- Developed a pilot project using *autonomous robots* for generating *digital twins* of construction facilities.

Key Publications

- **Minimum-Length Coverage Path Planning for Grid Environments with Approximation Guarantees**
IEEE Robotics and Automation Letters (RA-L), 2025
- **Approximate Environment Decompositions for Robot Coverage Planning using Submodular Set Cover**
IEEE Conference on Decision and Control (CDC), 2024
- **Anytime Replanning of Robot Coverage Paths for Partially Unknown Environments**
IEEE Transactions on Robotics (T-RO), 2024
- **Optimal Partitioning of Non-Convex Environments for Minimum Turn Coverage Planning**
IEEE Robotics and Automation Letters (RA-L), 2022

Projects and Competitions

BenchNPIN - Benchmarking Non-Prehensile Interactive Navigation Project Website

Reinforcement learning (RL) benchmarking tools for robots navigating among movable obstacles.

Roboracer Sim Racing Competition - CDC 2024 Project Code

Developed autonomous driving algorithms to complete the qualification stage of a racing competition.

RoVR the Explorer (HackED 2019 – 3rd Place Win) Github Link

VR-controlled ground robot with a point cloud stream to visualize its environment.

Paulie Blart Security Robot (SFHacks 2018 – 1st Place Win) Github Link

Autonomous security robot to detect intruders (using CNNs). Seen on Global News (link).

Technologies

Languages: Python 3, C++ (14/17), C#, Matlab, Objective-C, Java

Technologies: Robotics (ROS1/2, MQTT), ML tools (NumPy, Pytorch, scikit-learn), Simulation/AR (Unity, MuJoCo), Docker, Git